

AYUSH KUMAR

Full Stack Web Developer

Gorakhpur, Uttar Pradesh | +91 9335280820 | ayushmaurya496@gmail.com | github.com/ayushkumar1420 |
linkedin.com/in/ayush-kumar-83a969347 | ayush1420.vercel.app

PROFESSIONAL SUMMARY

Full Stack Web Developer building end-to-end features across responsive React and Tailwind CSS interfaces, Node.js/Express.js services, RESTful APIs, and real-time Socket.IO interactions. Skilled in JavaScript, TypeScript, MongoDB, Redis, authentication, failure handling, and full-stack debugging through hands-on web projects.

KEY PROJECTS

Seat Lock Engine

Node.js, Express.js, Redis, MongoDB, RESTful APIs

- Engineered a high-concurrency reservation backend to prevent double-booking when users request the same seat simultaneously.
- Implemented atomic Redis seat locks with TTL-based expiration to serialize competing requests and automatically release timed-out holds.
- Designed RESTful booking APIs and a PENDING lifecycle coordinating Redis locks with MongoDB persistence, including rollback when database writes fail.
- Built background expiry processing to release abandoned reservations and keep booking state and seat availability consistent.

Real-Time Chat Application

React.js, Node.js, Express.js, Socket.IO, MongoDB, Zustand

- Developed a responsive full-stack messaging application with React.js and Express.js, using Socket.IO for real-time bidirectional communication and presence updates.
- Designed RESTful APIs and MongoDB models for user accounts and conversations, securing sessions with JWT-based authentication.
- Integrated Cloudinary for profile-image uploads and media management, and used Zustand for predictable client-side state management.
- Debugged client-server integration and built mobile-friendly interaction flows for consistent behavior across desktop and mobile browsers.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Zustand/Redux, Responsive Web Design

Backend: Node.js, Express.js, RESTful API Design, Socket.IO, JWT Authentication, Error Handling

Databases: MongoDB, MySQL, Redis, Firebase, Supabase, Schema Design, CRUD Operations

Tools & Cloud: Git/GitHub, Postman, Vercel, Render, AWS, CI/CD, Debugging & Troubleshooting

EDUCATION

B.Tech in Information Technology, Buddha Institute of Technology, Gorakhpur | 2023 - 2027

Class XII (PCM), Sanskriti Public School, Gorakhpur | 2022

Class X, St. Xavier's School, Bhatpar Rani, Deoria | 2020

LEADERSHIP & ACTIVITIES

- Presented a peer-reviewed research paper at the NCET-CSIT National Conference (2026), communicating technical concepts to an academic audience.
- Collaborated with faculty members and student volunteers to organize a technical event focused on problem-solving and competitive programming.
- Balanced coursework with multiple independent full-stack projects and ongoing DevOps/AWS certification study, demonstrating ownership and self-direction.